**Duy Cao-Duc**

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## Personal Statement

My research lies at the intersection of 3D generation, understanding, and robotics. I develop efficient 3D generative models that produce assets for both creative applications and robotic simulation. My long-term goal is to advance 3D intelligence through 3D content generation and physical AI systems capable of understanding and interacting with the real world. I am determined to pursue a Ph.D. and contribute impactful research toward realizing this vision.

## Education

**FPT University** Hanoi, Vietnam
Bachelor in Artificial Intelligence. 2021 - 2025

- **CGPA:** 8.90/10.00 (3.63/4.00). **Valedictorian**, 2025 Information Technology graduating class.
- **Thesis:** Next-Gen Multi-modal 3D Synthesis Without Vector Quantization

## Experience

**Qualcomm AI Research** Hanoi, Vietnam
**AI Resident** – Supervisors: Dr. Phong Nguyen and Dr. Anh Tran 03/2025 - Present

- Developed **PixGS**, a single-stage pixel-space diffusion model for direct 3D Gaussian Splat generation, published at ECCV 2026.
- Quantized and deployed **PixGS** on Qualcomm XR/VR glasses.
- Developed **Next3DGen**, a reward-based post-training framework for text- and image-conditioned 3D generators, under review at ICLR 2027.
- Investigated efficient VLA models and retrieval-based long-term memory for real-world robotic deployment.

**VinAI Research** Hanoi, Vietnam
**AI Resident** – Supervisors: Dr. Phong Nguyen and Dr. Anh Tran 08/2024 - 03/2025

- Investigated multimodal 3D language models that jointly support 3D generation and understanding.
- Investigated feed-forward neural scene rendering and 3D reconstruction based on Gaussian and triangle splatting.

**AI Engineer** - Smart Edge Team 01/2024 - 08/2024

- Contributed to the development and deployment of face recognition, anomaly detection, and human tracking systems for city-scale surveillance.

## Publications

**Conference Proceedings**

1. **Cao Duy** and Phong Nguyen-Ha, "PixGS: Pixel-Space Diffusion for Direct 3D Gaussian Splat Generation," in *European Conference on Computer Vision (ECCV)*, 2026. [arXiv]
2. **Cao Duy**, Hong-Phuc Lai, Anh Tran, and Phong Nguyen-Ha, "The Next Step in 3D Generation Post-Training," under review at the *International Conference on Learning Representations (ICLR)*, 2027. [PDF]

## Awards and Achievements

**Valedictorian**, Information Technology graduating class (1st among 1,000+ graduates) 2025
**Full Four-Year Scholarship Recipient** 2021–2025
**Inventor on Two Pending U.S. Patent Applications**: *PixGS* and *Next3DGen* 2026
**Vietnam AI Talent Nominee**, nominated by VnExpress 2025
**Third Prize**, FPT Edu Digital Race 2023